

Total No. of Questions : 4]

SEAT No. :

PC403

[6359]-523

[Total No. of Pages :2

S.E. (Computer Engineering) (Insem)
DIGITAL ELECTRONICS AND LOGIC DESIGN
(2019 Pattern) (Semester- III) (210245)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

Q1) a) Solve the following equation using K-map minimization techniques. Also draw the diagram using basic gates

$$F(A,B,C,D)=\Sigma m(2,7,8,10,11,13,15) \quad [5]$$

b) Minimize the function in POS form and also draw the diagram

$$F(A,B,C,D)=\pi M(0,1,3,5,6,7,9,10,15) \quad [5]$$

c) How will you implement full adder using half adder? Explain the circuit diagram. [5]

OR

Q2) a) Represent the following decimal numbers in [5]

- i) Sign magnitude form
- ii) Sign 1's complement form
- iii) Sign 2's complement form

- 1) +14
- 2) +27
- 3) -17
- 4) +45

b) Minimize the following logic function using K-map and realize using logic gates. Draw the diagram $F(A,B,C,D)=\Sigma m(1,3,7,11,15)+d(0,2,5)$ [5]

c) Simplify logic equation using Quine Mc Cluskey minimization Technique $Y(A,B,C,D)=\Sigma m(0,1,3,7,8,9,11,15)$ [5]

P.T.O.

Q3) a) Design and Implement 4 bit Binary to Gray code converter using logic gate. [5]

b) Design single digit BCD adder using 4 Bit binary adder IC 7483. [5]

c) Implement the expression using a 8:1 multiplexer

$$F(A,B,C,D)=\Sigma m(0,2,3,6,8,9,12,14) \quad [5]$$

OR

Q4) a) Write a short note on look ahead carry generator also explain its importance. [5]

b) Design 2-bit magnitude comparator using logic gates. [5]

c) Design 3 bit parity generator circuit using odd parity bit. [5]